Lecture 2 Exercise Solutions

1. Prove that in any group for any elements $h, x, y \in G$ we have $h(xy)h^{-1} = (hxh^{-1})(hyh^{-1})$.

   $(hxh^{-1})(hyh^{-1}) =$
   $(hx)[h^{-1}(hyh^{-1})] =$
   $(hx)[(h^{-1}h)(yh^{-1})] =$
   $(hx)[e(yh^{-1})] =$
   $(hx)(yh^{-1}) =$
   $h(xy)h^{-1}$

   Note: We only use the associative property in this proof and because we didn't state that this is an Abelian Group, we can't use commutative property to reorder the factors.

2. Make an addition table for Z_4.

   Z_4 has four blocks, or equivalence classes, $Z_4 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$, so that operation addition, mod 4, is given by the table

	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{0}$	$\bar{1}$	$\bar{2}$

3. Why isn't $\{\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$ a group under addition or multiplication, mod 6?

   The identity element under addition is $\bar{0}$ but $\bar{0}$ is not in the set. Under multiplication $\bar{2} \cdot \bar{3} = \bar{0}$, but, again, $\bar{0}$ is not in the set.

4. Prove by induction that in an Abelian group, G, we have $(gh)^n = g^n h^n$ for all $g, h \in G$ and all n.

   We'll prove the statement by induction. First lets' check if $(gh)^2 = g^2 h^2$:

$(gh)^2 =$
$(gh)(gh) =$
$g[h(gh)] =$
$g[h(hg)] =$
$g(h^2 g) =$
$g(gh^2) =$
$g^2 h^2$

Now, we make an inductive assumption that $(gh)^n = g^n h^n$ for all numbers from 1 to n and prove that $(gh)^{n+1} = g^{n+1} h^{n+1}$. Note that we prove this in terms of $(gh)^n = g^n h^n$:

$(gh)^{n+1} =$
$(gh)^n (gh) =$
$(g^n h^n)(gh) =$
$g^n [h^n (gh)] =$
$g^n [h^n (hg)] =$
$g^n (h^{n+1} g) =$
$g^n (gh^{n+1}) =$
$g^{n+1} h^{n+1}$

5. Prove that $(gh)^{-1} = h^{-1} g^{-1}$.

$(gh)(h^{-1} g^{-1}) =$
$g(hh^{-1}) g^{-1} =$
$geg^{-1} =$
$gg^{-1} =$
e

So, obviously, $h^{-1} g^{-1}$ is the inverse of gh. But, by definition, so is $(gh)^{-1}$. Therefore they must be equal.

6. Prove that the intersection of two subgroups H and K of a group G is a subgroup of G.

A group must be associative under its operation, have an identity element, be closed, and every element must have an inverse. Let's show, or prove, each in turn:

i. Associativity

Clearly, if the operation of amongst the elements in G is associative, then it must be associative in any subgroup.

ii. Identity

$e \in H \wedge e \in H \Rightarrow e \in H \cap K$, so the intersection has the identity element.

iii. Closed

If $x, y \in H \bigcap K \Rightarrow x, y \in H \wedge x, y \in K$. Since $x, y \in H$, then since H is a group, then $x * y \in H$ under closure. Similarly, since $x, y \in K$ and K is a group, then $x * y \in K$ under closure. Now,
$x * y \in H \wedge x * y \in K \Rightarrow x * y \in H \bigcap K$, so the intersection is closed.

iv. Inverse

If $x \in H$ then since H is a group, then $x^{-1} \in H$ under the inverse property. Similarly, if $x \in K$ and since K is a group, then $x^{-1} \in K$ under the inverse property. Now, $x^{-1} \in H \wedge x^{-1} \in K \Rightarrow x^{-1} \in H \bigcap K$, so every element has an inverse in the intersection.

7. Show that in the Abelian group G, the set $H = \left\{ g \in G \middle| g^n = e, \forall n > 0 \right\}$ is a subgroup.

A group must be associative under its operation, have an identity element, be closed, and every element must have an inverse. Let's show, or prove, each in turn:

i. Associativity

Clearly, if the operation of amongst the elements in G is associative, then it must be associative in any subgroup.

ii. Identity

If $g = e \wedge g \in H \Rightarrow e \in H.$ Clearly, $e^n = e$.

iii. Closed

Let $x \in H$, then $x^n = e$, and let $y \in H$, then $y^n = e$. Then,
$(xy)^n = x^n y^n = e * e = e$.

iv. Inverse

$x \in H \wedge H \subseteq G \Rightarrow x \in G$. Since, G is a group and $x \in G$ there must be an inverse, $x^{-1} \in G$. It exists as $x \in G$. It will also belong to H if $(x^{-1})^n = e$. But $x^n * (x^{-1})^n = (x * x^{-1})^n = e^n = e$ and $x^n = e$, thus $(x^{-1})^n = e$.

8. Find all 5 of the distinct subgroups of Z_{16}.

   i. $\{\bar{0}\}$
   ii. $\{\bar{0}, \bar{8}\}$
   iii. $\{\bar{0}, \bar{4}, \bar{8}, \overline{12}\}$
   iv. $\{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \overline{10}, \overline{12}, \overline{14}\}$
   v. Z_{16}

9. Find all 8 of the distinct subgroups of the group Z_{24}.

   i. $\{\bar{0}\}$
   ii. $\{\bar{0}, \overline{12}\}$
   iii. $\overline{\{0}, \bar{8}, \overline{16}\}$
   iv. $\{\bar{0}, \bar{4}, \bar{8}, \overline{12}, \overline{16}, \overline{20}\}$
   v. $\{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \overline{10}, \overline{12}, \overline{14}, \overline{16}, \overline{18}, \overline{20}, \overline{22}\}$
   vi. $\overline{\{0}, \bar{6}, \overline{12}, \overline{18}\}$
   vii. $\{\bar{0}, \bar{3}, \bar{6}, \bar{9}, \overline{12}, \overline{15}, \overline{18}, \overline{21}\}$
   viii. Z_{24}

   Observe that the number of elements in the subgroups divides the number of elements in the group. The divisors of 24 are $1,2,3,4,6,8,12,24$ so the number of subgroups equals the number of divisors.

10. List all elements of $Z_{30}^{\times}$ and prove that they form a group.

    You can derive the following members using the Euclidean algorithm. Only integers that are relatively prime to 30 are in the group. That is, $\gcd(x, 30) = 1$ where $x \in Z_{30}^{\times}$.

$Z_{30}^{\times} = \{\bar{1}, \bar{7}, \overline{11}, \overline{13}, \overline{17}, \overline{19}, \overline{23}, \overline{29}\}$

In addition, a group has four additional properties:

i. Associativity – Obviously, integer multiplication is associative.
ii. Identity – 1 is the identity element in this group.
iii. Inverse – Every element must have an inverse. Clearly, if all these elements, x, are relatively prime to 30 then the Extended Euclidean Algorithm allows us to calculate $\gcd(x, 30) = a \cdot x + b \cdot 30 = 1$. We can rearrange this equation as follows: $a \cdot x = (-b) \cdot 30 + 1$. Clearly, here $-b$ is just another constant and remember that this can be recast as $a \cdot x \ \% \ 30 = 1$. So every x has an inverse a.
iv. Closure – We proved in Lecture 1, Proposition 63 proves that if two elements, x and y are relatively prime to 30, then they're product is relatively prime to 30.

So, we have all the have the properties of a group, so are elements of the group. Also, it is obvious that since integer multiplication is commutative, this is also an Abelian group.

11. Check that the collection of matrices g of the form $g = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ is a subgroup.

A group must be associative under its operation, have an identity element, be closed, and every element must have an inverse. Let's show, or prove, each in turn:

i. Associativity

Matrix multiplication is associative.

ii. Identity

$$e = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

iii. Closed

$$\begin{pmatrix} a_1 & 0 \\ 0 & b_1 \end{pmatrix} \times \begin{pmatrix} a_2 & 0 \\ 0 & b_2 \end{pmatrix} = \begin{pmatrix} a_1 a_2 & 0 \\ 0 & b_1 b_2 \end{pmatrix}$$

iv. Inverse

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \times \begin{pmatrix} \frac{1}{a} & 0 \\ 0 & \frac{1}{b} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = e$$

12. Show that a group of order 101 has no subgroups except $\{e\}$ and itself.

    If this group had a subgroup H, then $|H|$ would divide 101, but 101 is a prime number, so there are only two subgroups, $\{e\}$ and the group itself.

13. Show that a group of order 1009 has no subgroups except for $\{e\}$ and itself.

    If this group had a subgroup H, then $|H|$ would divide 1009, but 1009 is a prime number, so there are only two subgroups, $\{e\}$ and the group itself.

14. For group G, where $g, h \in G$, show that if $|\langle g \rangle| = 30$ the and $|\langle h \rangle| = 77$, then $|\langle g \rangle| \cap |\langle h \rangle| = \{e\}$.

    The intersection of any two subgroups is a subgroup of each of them, so the order of the intersection should a divisor of the sizes of the two subgroups. But 30 and 77 are relatively prime, so the only divisor is 1. That is, the size of the intersection is 1 and that means that the intersection has only the identity element in it.

15. Prove that a group element and its inverse have the same order.

    Let the order of $|\langle g \rangle| = n$ for $g \in G$. That is, $g^n = e$. Let g^l be the inverse of g. Then, $g^n * (g^l)^n = (g * g^l)^n = e$. But, $g^n = e$, so $e * (g^l)^n = e \Rightarrow (g^l)^n = e$. Thus, the inverse, g^l, has the same order as g.

16. Check that $Z_8^\times$ cannot be generated by a single element.

    $Z_8^\times = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$

    $\forall n \left(\bar{1}^n = \bar{1} \right)$, so $\langle \bar{1} \rangle = \{\bar{1}\}$

    $\bar{3}^1 = \bar{3}$ and $\bar{3}^2 = \bar{1}$, so $\langle \bar{3} \rangle = \{\bar{1}, \bar{3}\}$

$\bar{5}^1 = \bar{5}$ and $\bar{5}^2 = \bar{1}$, so $\langle \bar{5} \rangle = \{\bar{1}, \bar{5}\}$

$\bar{7}^1 = \bar{7}$ and $\bar{7}^2 = \bar{1}$ so $\langle \bar{7} \rangle = \{\bar{1}, \bar{7}\}$